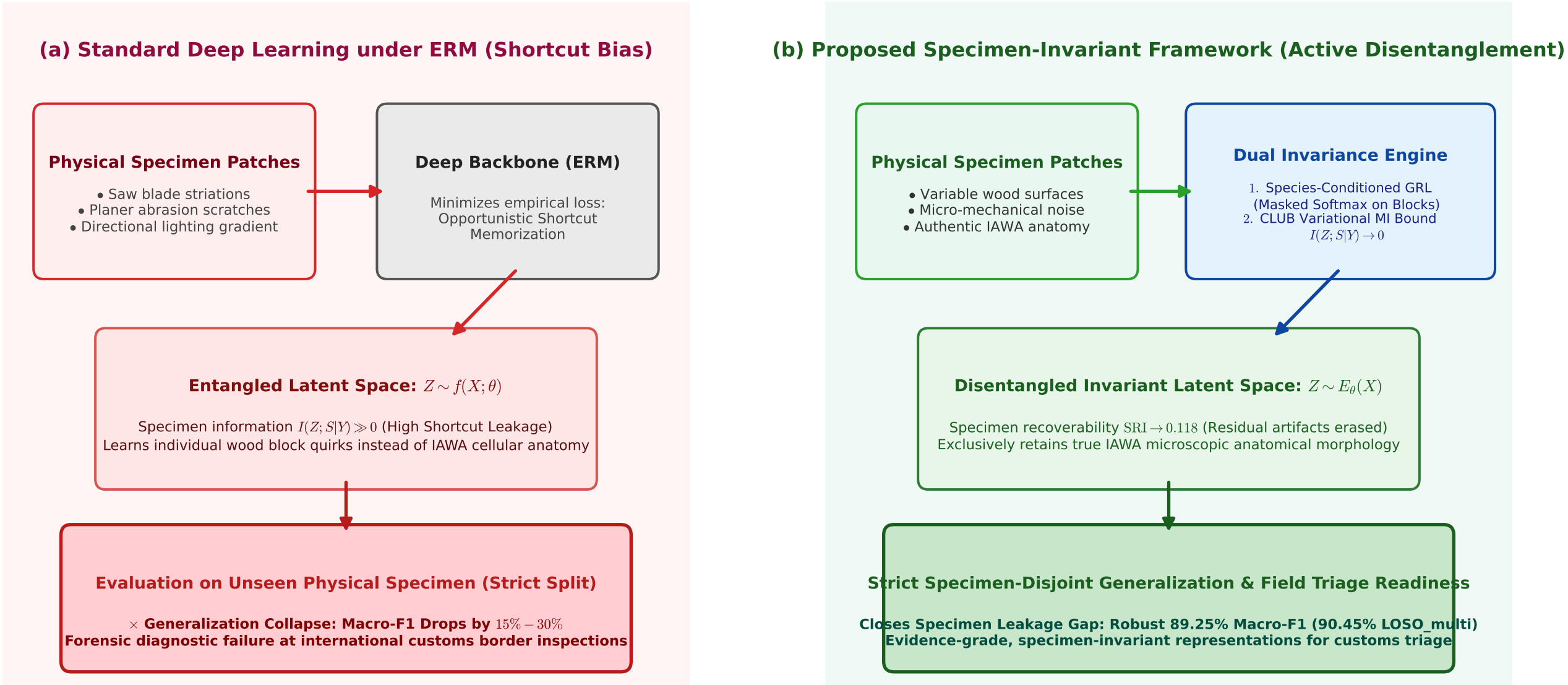

(a) Standard Deep Learning under ERM (Shortcut Bias)
Physical Specimen Patches
• Saw blade striations
• Planer abrasion scratches
• Directional lighting gradient
Deep Backbone (ERM)
Minimizes empirical loss:
Opportunistic Shortcut
Memorization
Entangled Latent Space: $Z \sim f(X;\theta)$
Specimen information $I(Z;S|Y) \gg 0$ (High Shortcut Leakage)
Learns individual wood block quirks instead of IAWA cellular anatomy
Evaluation on Unseen Physical Specimen (Strict Split)
× Generalization Collapse: Macro-F1 Drops by $15\% - 30\%$
Forensic diagnostic failure at international customs border inspections
(b) Proposed Specimen-Invariant Framework (Active Disentanglement)
Physical Specimen Patches
• Variable wood surfaces
• Micro-mechanical noise
• Authentic IAWA anatomy
Dual Invariance Engine
1. Species-Conditioned GRL
(Masked Softmax on Blocks)
2. CLUB Variational MI Bound
$I(Z;S|Y) \to 0$
Disentangled Invariant Latent Space: $Z \sim E_\theta(X)$
Specimen recoverability $\mathrm{SRI} \to 0.118$ (Residual artifacts erased)
Exclusively retains true IAWA microscopic anatomical morphology
Strict Specimen-Disjoint Generalization & Field Triage Readiness
Closes Specimen Leakage Gap: Robust 89.25% Macro-F1 (90.45% LOSO_multi)
Evidence-grade, specimen-invariant representations for customs triage